

VTQAGen: BART-based Generative Model For Visual Text Question Answering

Haoru Chen*
Key Laboratory of Data Engineering
and Visual Computing, Chongqing
University of Posts and
Telecommunication

Tianjiao Wan*
National University of Defense
Technology
Changsha, Hunan, China

Zhimin Lin*
MindRank AI Ltd.,
Hangzhou, China

Kele Xu
National University of Defense
Technology
Changsha, Hunan, China

Jin Wang
Key Laboratory of Data Engineering
and Visual Computing, Chongqing
University of Posts and
Telecommunication

Huaimin Wang
National University of Defense
Technology
Changsha, Hunan, China

ABSTRACT

Visual Text Question Answering (VTQA) is a challenging task that requires answering questions pertaining to visual content by combining image understanding and language comprehension. The main objective is to develop models that can accurately provide relevant answers based on complementary information from both images and text, as well as the semantic meaning of the question. Despite ongoing efforts, the VTQA task presents several challenges, including multimedia alignment, multi-step cross-media reasoning, and handling open-ended questions. This paper introduces a novel generative framework called VTQAGen, which leverages a Multi-modal Attention Layer to combine image-text pairs and question inputs, as well as a BART-based model for reasoning and entity extraction from both images and text. The framework incorporates a step-based ensemble method to enhance model performance and generalization ability. VTQAGen utilizes an encoder-decoder generative model based on BART. Faster R-CNN is employed to extract visual regions of interest, while BART's encoder is modified to handle multi-modal interaction. The decoder stage utilizes the shift-predict approach and introduces step-based logits fusion to improve stability and accuracy. In the experiments, the proposed VTQAGen demonstrates superior performance on the testing set, securing second place in the ACM Multimedia Visual Text Question Answer Challenge.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning algorithms**;
Natural language processing.

*First three authors contributed equally to this research.
The corresponding author is Kele Xu, xukele@nudt.edu.cn

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0108-5/23/10...\$15.00
<https://doi.org/10.1145/3581783.3612844>

KEYWORDS

Visual Text Question Answering, Multi-modal Attention, Cross-media Reasoning

ACM Reference Format:

Haoru Chen, Tianjiao Wan*, Zhimin Lin*, Kele Xu, Jin Wang, and Huaimin Wang. 2023. VTQAGen: BART-based Generative Model For Visual Text Question Answering. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3581783.3612844>

1 INTRODUCTION

Visual question answering (VQA) is a significant task situated at the intersection of computer vision and natural language processing [3, 15, 18]. Its purpose is to address the challenge of answering questions related to visual content by effectively combining image understanding and language comprehension [6, 21]. The primary objective of VQA is to develop models capable of providing relevant and accurate answers to textual questions based on the visual information contained in an image [1, 39, 46]. To achieve this goal, the model must comprehend both the visual features present in the image and the underlying semantic meaning embedded within the question, enabling the generation of appropriate responses [20, 23, 37, 44].

Despite sustained efforts in the field of Visual Question Answering (VQA), it is crucial to acknowledge that most VQA research to date has primarily concentrated on extracting information solely from single image data [14, 17, 33]. However, this approach fails to align with the human cognitive process, as humans tend to integrate information from multiple modalities when answering questions [2, 9, 10, 41, 48].

To bridge this gap, several recent endeavors have aimed to incorporate additional modalities, such as text, into the Visual Question Answering (VQA) task [34, 36]. The task of Visual Text Question Answering (VTQA) presents several challenges. Firstly, aligning multimedia representations of the same entity in images and text has been a task in the field of multimedia, requiring models to relate information from multiple sources. Temporal and cross-modality regulation serve as organic supervision for large-scale models [7, 12, 13]. In the context of VTQA, images are closely connected to the text, which

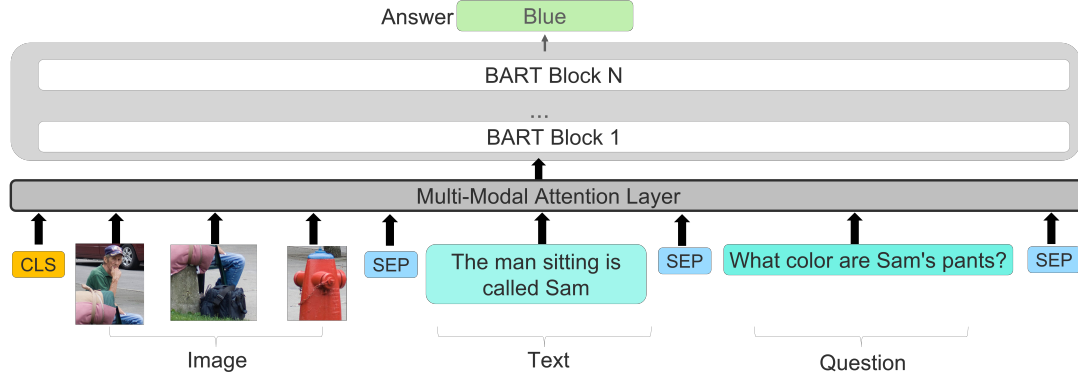


Figure 1: The VTQAGen framework tackles the multi-modal visual text question-answering problem through an encoder-decoder architecture based on BART. The framework utilizes Faster-RCNN to extract the regions of interest (ROI) in images and combines text embeddings from question and image description texts. The encoder facilitates multi-modal interaction and fusion, while the decoder employs a shift-predict strategy for token prediction.

may contain semantic entities referred to in the questions. Therefore, it is crucial for the proposed framework to learn not only to identify entities within each modality but also to align multimedia representations of the same entity across image and text.

Secondly, due to the complexity of the VTQA, multi-step cross-media reasoning is required between image-text pairs and questions. The initial step involves analyzing VTQA questions and identifying key entities based on the questions. Grounding the key entities between the image and text, the framework should predict open-ended answers by considering question entities and aligned entities, which play a crucial role in connecting the information presented in the image-text pair and the question text [22, 24, 27, 30].

Thirdly, unlike closed-ended questions where the answer is limited to predefined options, open-ended questions require more human-like responses [16, 35, 38, 42, 46]. However, the challenge with open-ended questions is that they can lead to lengthy or rambling responses. Without the constraint of predefined answer choices, respondents may provide overly detailed or irrelevant information [8, 32, 45, 47, 49].

In this work, we propose a novel framework for the VTQA task, named VTQAGen, to address the aforementioned challenges. First, we introduce a Multi-modal Attention Layer to learn fused multi-modal features using input from image-text pairs and questions [5, 26, 40, 43]. Next, we employ the BART model to implement multi-step reasoning and identify entities in images and text related to the given question [4, 19, 29]. Finally, to enhance the performance and generalization of our VTQAGen model, we introduce a step-based ensemble method to train five VTQAGen models on various subsets and aggregate answers at the token level.

2 METHODOLOGY

2.1 Overall Framework

VTQAGen is built upon an encoder-decoder generative model, leveraging BART as the main model backbone (as shown in Figure 1). We introduce modifications to the encoder component to fuse visual

and textual interaction information, thereby generating encoder-hidden representations that capture context information. In the decoder stage, we employ the shift-predict mechanism where the previous token is used to predict the next token, enabling the generation of open-ended answers. Given the multi-modal nature of the VTQA task, which involves both visual and textual information, we introduce the Multi-modal Attention Layer to effectively learn aligned representations of entities present in both the visual image and its corresponding descriptive text. Accurate generation of answers relies on a deep comprehension of the complementary information from both modalities to identify the corresponding entities referenced in the question. Inspired by the powerful learning capabilities of the BART model, we leverage it to facilitate the multi-step cross-media reasoning process between image-text pairs and questions. Fused multimodal embeddings are used as input, allowing the framework to implicitly reason over the grounding between the image and text modalities and identify relevant entities mentioned in the given question. The ultimate goal of our framework is to generate accurate answers by effectively leveraging the interplay between different modalities and reasoning about their relationships.

2.2 Multi-modal Attention Layer

The key challenges in VTQA lie in aligning multimedia representations of the same entity in image and text, as well as identifying the entities in the image and text that are mentioned in the question. To overcome this challenge, we design a simple yet effective Multi-modal Attention Layer. The Visual and Textual Embedder are introduced to extract embeddings from regions of the image, textual tokens from the text, and textual tokens from the question, respectively (as shown in Figure 2). Then these embeddings are concatenated and passed into the Multi-modal Attention Layer. To be specific, the visual embeddings are generated by extracting features of Region of Interest using Faster-RCNN; while the textual embeddings are generated by tokenizing input sentences from both the text and questions into WordPieces, following the approach employed in BERT [11]. Multi-modal Attention layer treats the embeddings as queries, keys,

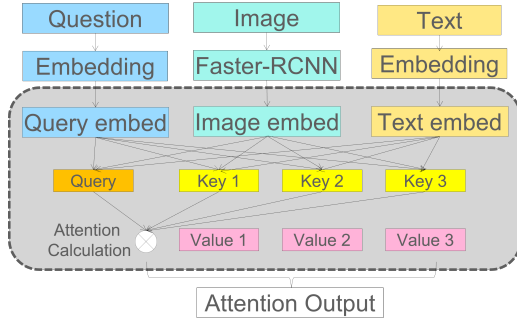


Figure 2: Multi-modal Attention Layer.

and values, similar to a standard transformer block, regardless of their origin from textual or visual content. By performing attention over the entire input sequence, this design allows for learning fused multimodal features that capture the relationships between visual regions, the associated descriptive of the text, and textual tokens of the question, which paves the way for subsequent reasoning.

2.3 Bidirectional and Auto-Regressive Transformers (BART)

In the BART model [25], the encoder processes the input sequence, capturing contextual information from both the left and right contexts, thereby enabling bidirectional information flow. This helps in capturing dependencies between words more effectively. On the other hand, the decoder autoregressively generates the output sequence, attending only to previously generated tokens during the decoding process. This approach ensures that the model adheres to the autoregressive property, maintaining the left-to-right generation constraint. Given the remarkable success of the BART model in sequence generation tasks, we are highly motivated to explore its applicability in addressing the novel challenge of VTQA task. Unlike models for multi-classification task, the BART model excels at outputting an impressive answer sequence.

2.4 Step-Based Ensemble for Multi-modal Reason

To enhance the robustness and accuracy of open-ended answers in our VTQAGen model, we have incorporated the Bootstrap aggregating (Bagging) algorithm, an effective ensemble learning method. Bagging creates multiple subsets of the original dataset through bootstrapping, and trains multiple models on these subsets, sampled with replacement from the training data. The final prediction is obtained by aggregating the predictions of these models, typically through averaging. This ensemble approach significantly improves the reliability of the final output and reduces the average generalization error, leading to enhanced prediction performance.

In our method, we ensemble five VTQAGen models trained on diverse subsets of the data, capturing various patterns and improving the overall quality of the generated text. However, as VTQAGen generates text for the VTQA task, simple averaging of the models' outputs cannot be directly applied to obtain the final result. To overcome this challenge, we propose a novel step-based ensemble

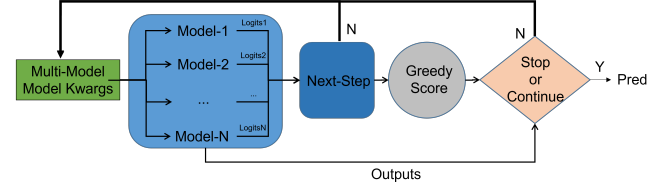


Figure 3: Step-Based Ensemble.

method that effectively aggregates answers from multiple models (as shown in Figure 3). During the prediction process of the generative model, where the probability distribution of the next token is estimated, we average the probability outputs from each model for each token. Then, we use the Greedy search strategy to select the most probable word as the next token at each step. This token-based ensemble approach leverages the collective knowledge and diversity of the individual models, resulting in more accurate answers for the VTQA task. By harnessing the strengths of each model and mitigating potential weaknesses, this method improves performance and provides more reliable open-ended answers.

3 EXPERIMENTAL RESULTS

3.1 Datasets

We utilize the VTQA dataset [7] to train our model, which is specifically designed for the open-ended multimedia VTQA challenge. The dataset consists of 10,124 image-text pairs, accompanied by 23,781 questions. To facilitate comprehensive evaluation, the dataset is divided into four subsets: training, validation, Dev, and test. Each image is exclusively present in one split, ensuring data integrity. The sample counts for each split are as follows: 11,312 for training, 1,245 for validation, 2,189 for Dev, and 9,035 for test. The images in the VTQA dataset are sourced from the real-world MSCOCO dataset [28], which is known for its rich contextual information.

3.2 Implementations details

To train the VTQAGen framework on the VTQA Chinese dataset, we utilize the pre-trained BART-base-Chinese model. This model is specifically designed for handling Chinese data and is built upon the BART architecture. In the context of our VTQA task, we fine-tune this model on the multimodal data from the Chinese version of the VTQA dataset for 20 epochs. The dimensions of the image region features, input text features, and input question features are set to 2048, 768, and 768, respectively. The fused multimodal features have a dimensionality of 768. The BART-base-Chinese model consists of 12 blocks. To ensure efficient training, we use a batch size of 11. During the training phase, we employ the AdamW optimizer [31] with default settings. We also utilize a Cosine learning rate scheduler and apply a 20-epoch cosine warm-up. The learning rate is initialized to $1e-4$, and a weight decay of 0.01 is applied. These settings remain consistent during the validation and test phases to ensure reproducibility.

For the English VTQA dataset, we utilize the pre-trained BART-base model, which is trained on a large corpus of English text. To complete the VTQA task on this dataset, we fine-tune the model using multimodal data from the Chinese version of the VTQA dataset

Table 1: Experimental results with different settings. (Red: Best; Blue: Second Best)

Model Setting			Tricks		Chinese				English			
Cross Modal	Decoder	Visual Encoder	Ensemble	Fine-tune	Dev		Test-Dev		Dev		Test-Dev	
					EM	F1	EM	F1	EM	F1	EM	F1
Baseline					0.527	0.619	0.3892	0.5997	0.5923	0.6723	0.4437	0.5831
Single Stream	BART	VIT	Single Model	None	0.5532	0.6421	0.4367	0.6321	0.6319	0.7042	0.4825	0.6297
Double Streams	BART	Faster-RCNN	Single Model	None	0.4072	0.5723	0.2852	0.5492	0.4513	0.472	0.3123	0.4696
Double Streams	BART	VIT	Single Model	None	0.1277	0.323	0.1824	0.3023	0.1732	0.3524	0.1328	0.2482
Single Stream	BART	Faster-RCNN	Single Model	None	0.6730	0.7852	0.5491	0.7035	0.7172	0.8192	0.5793	0.7042
Single Stream	T5	Faster-RCNN	Single Model	None	0.1293	0.1421	0.2183	0.3291	0.1724	0.1502	0.2481	0.3831
Single Stream	BART	Faster-RCNN	Single Model	EMA	0.6827	0.7948	0.5487	0.7001	0.7220	0.8224	0.5770	0.7064
Single Stream	BART	Faster-RCNN	Single Model	EMA, FGM	0.6875	0.7953	0.5560	0.7109	0.7231	0.8250	0.5811	0.7332
Single Stream	BART	Faster-RCNN	Single Model	R-Drop	0.6425	0.7554	0.4673	0.6091	0.7123	0.8142	0.5651	0.7050
Single Stream	BART	Faster-RCNN	Bagging	EMA, FGM	-	-	0.5838	0.7816	-	-	0.5975	0.7194

for 20 epochs. The parameter settings remain the same as those used in the training of the Chinese task.

3.3 Experimental Results

3.3.1 Quantitative Results. Table 1 presents the experimental results of the VTQAGen model under different settings. The table provides a comprehensive overview of various model configurations, training tricks, and the achieved performance metrics (EM and F1) on both the Chinese and English VTQA datasets. To analyze the impact of different framework designs, especially in handling multi-modal information, we compare the performance of our VTQAGen with the Double Streams Cross Attention Framework. In this framework, the key-value pairs from the question embedding are passed as inputs to the multi-head attention blocks of both the image and the text, integrating the question information into each modality. The Double Streams Cross Attention Framework includes 2 multi-head attention blocks and 12 BART blocks, which is consistent with the setup in our VTQAGen. However, despite its more complex and intuitive structure, it significantly underperforms compared to VTQAGen, as shown in Table 1.

The “Baseline” configuration serves as the reference point, achieving an EM score of 0.527 and an F1 score of 0.619 on the dev set for the Chinese dataset. These baseline scores provide a benchmark for evaluating the effectiveness of subsequent model variations. Next, we analyze the performance of different model settings. In the “Single Stream” category, using the BART decoder with the Faster-RCNN visual encoder yields improved results compared to using the VIT visual encoder. For example, the BART-Faster-RCNN configuration achieves an EM score of 0.6730 and an F1 score of 0.7852 on the dev set for the Chinese dataset, while the BART-VIT configuration achieves lower scores (0.5532 for EM and 0.6421 for F1). Similar trends are observed for the English dataset.

Additionally, when employing the “Double Streams” approach with the BART decoder and Faster-RCNN visual encoder, lower scores are achieved compared to the Single Stream configurations. For instance, the Double Streams (BART-Faster-RCNN) configuration achieves an EM score of 0.4072 and an F1 score of 0.5723 on the dev set for the Chinese dataset. In comparison, the Single Stream (BART-Faster-RCNN) configuration achieves higher scores (0.673 for EM and 0.7852 for F1). This indicates that the Single Stream approach outperforms the Double Streams configuration.

3.3.2 Ablation study. The table also provides insights into the ablation study, where specific tricks or modifications are applied to the models. For instance, applying Exponential Moving Average (EMA) and Fast Gradient Method (FGM) techniques to the Single Stream (BART-Faster-RCNN) model improves the performance. The EM score increases from 0.673 to 0.6875, and the F1 score increases from 0.7852 to 0.7953 on the dev set for the Chinese dataset.

Moreover, employing the Bagging technique in the Single Stream (BART-Faster-RCNN) model with EMA and FGM further enhances the results. Although the specific EM and F1 scores are not available on the dev set for the Bagging configuration, it achieves a higher EM score of 0.5838 on the test-dev set for the Chinese dataset compared to the baseline score of 0.3892. This suggests that Bagging effectively improves the model’s performance.

The quantitative analysis reveals that different model configurations significantly influence the performance of the VTQAGen model. Single Stream approaches with specific combinations of decoders and visual encoders generally achieve better results, while Double Streams configurations exhibit lower scores. Furthermore, applying techniques like EMA, FGM, and Bagging enhances the model’s performance, demonstrating the importance of these tricks in improving the accuracy of generated answers in VTQA.

4 ACKNOWLEDGMENTS

This work is supported by the major Science and Technology Innovation 2030 “New Generation Artificial Intelligence” project 2021ZD0112904.

5 CONCLUSION

In this paper, we introduce the VTQAGen framework, which utilizes a BART-based encoder-decoder architecture for reasoning and generating answers. It leverages multi-modal attention to fuse information from image-text pairs and questions, enabling alignment of multimedia representations. To enhance model performance, token-based ensemble training and a step-based logits fusion technique are introduced. The VTQAGen model achieves superior accuracy in the VTQA Challenge. In conclusion, the VTQAGen framework demonstrates advancements in multi-modal visual text question answering, showing promising results in addressing complex challenges and enhancing the interaction between humans and machines in a visually-driven world.

REFERENCES

- [1] Aishwarya Agrawal, Jiasen Lu, Stanislaw Antol, Margaret Mitchell, Dhruv Batra, C Lawrence Zitnick, and Devi Parikh. 2017. VQA: Visual Question Answering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 21–29.
- [2] Peter Anderson, Xiaodong He, Chris Buehler, Damien Teney, Mark Johnson, Stephen Gould, and Lei Zhang. 2018. Bottom-Up and Top-Down Attention for Image Captioning and Visual Question Answering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 6077–6086.
- [3] Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C Lawrence Zitnick, and Devi Parikh. 2017. VQA: Visual Question Answering. *International Journal of Computer Vision* 123, 1 (2017), 4–31.
- [4] Ali Furkan Biten, Ron Litman, Yusheng Xie, Srikar Appalaraju, and R Manmatha. 2022. Latr: Layout-aware transformer for scene-text vqa. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 16548–16558.
- [5] Ali Furkan Biten, Ruben Tito, Andres Mafra, Lluís Gomez, Marçal Rusinol, Minesh Mathew, CV Jawahar, Ernest Valveny, and Dimosthenis Karatzas. 2019. Icdar 2019 competition on scene text visual question answering. In *2019 International Conference on Document Analysis and Recognition (ICDAR)*. IEEE, 1563–1570.
- [6] Ali Furkan Biten, Ruben Tito, Andres Mafra, Lluís Gomez, Marçal Rusinol, Ernest Valveny, CV Jawahar, and Dimosthenis Karatzas. 2019. Scene text visual question answering. In *Proceedings of the IEEE/CVF international conference on computer vision*. 4291–4301.
- [7] Kang Chen and Xiangqian Wu. 2023. VTQA: Visual Text Question Answering via Entity Alignment and Cross-Media Reasoning. *arXiv preprint arXiv:2303.02635* (2023).
- [8] Xun Chen, Lin Ma, Wei Zhang, Hongxun Yang, and Liang Zheng. 2021. Multimodal Word-Wise Attention for Visual Question Answering. In *International Joint Conference on Artificial Intelligence (IJCAI)*. IJCAI, 1805–1812.
- [9] Yen-Chun Chen, Zhe Gan, Linjie Li, Licheng Yu, Jingjing Li, Yu Cheng, and Jingjing Liu. 2021. UNITER++: A Hierarchical Pretrained Language Model for Unified Image and Text Representation. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 10536–10546.
- [10] Yen-Chun Chen, Linjie Li, Licheng Yu, Ahmed El Kholy, Faisal Ahmed, Zhe Gan, Yu Cheng, and Jingjing Liu. 2020. UNITER: Universal Image-Text Representation Learning. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 13194–13204.
- [11] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [12] Shiv Ram Dubey. 2021. A decade survey of content based image retrieval using deep learning. *IEEE Transactions on Circuits and Systems for Video Technology* 32, 5 (2021), 2687–2704.
- [13] Christoph Feichtenhofer, Haoqi Fan, Jitendra Malik, and Kaiming He. 2019. Slow-fast networks for video recognition. In *Proceedings of the IEEE/CVF international conference on computer vision*. 6202–6211.
- [14] Peng Gao, Qi Mao, Yufeng Zhou, Zhiheng Huang, Lei Wang, and Wei Xu. 2015. Are You Talking to a Machine? Dataset and Methods for Multilingual Image Question. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2296–2304.
- [15] Yudong Guo, Ziping Huang, Peng Li, Xiaobo Zhou, Houjing Wang, and Liang Lin. 2021. LMVR: Learning Multimodal Vision-Language Representation via Cross-Modal Relation Modeling. *arXiv preprint arXiv:2104.10821* (2021).
- [16] Darryl Hannan, Akshay Jain, and Mohit Bansal. 2020. ManyModalqa: Modality disambiguation and qa over diverse inputs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34. 7879–7886.
- [17] Drew A Hudson and Christopher D Manning. 2018. Compositional Attention Networks for Machine Reasoning. *arXiv preprint arXiv:1803.03067* (2018).
- [18] Drew A Hudson and Christopher D Manning. 2019. GQA: A New Dataset for Real-World Visual Reasoning and Compositional Question Answering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 6700–6709.
- [19] Zan-Xia Jin, Heran Wu, Chun Yang, Fang Zhou, Jingyan Qin, Lei Xiao, and Xu-Cheng Yin. 2021. Ruart: A novel text-centered solution for text-based visual question answering. *IEEE Transactions on Multimedia* (2021).
- [20] Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C Lawrence Zitnick, and Ross Girshick. 2017. CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2901–2910.
- [21] Kushal Kafle and Christopher Kanan. 2017. An analysis of visual question answering algorithms. In *Proceedings of the IEEE international conference on computer vision*. 1965–1973.
- [22] Jinsoo Kim and Gunhee Lee. 2020. Dense Relational Captioning: Triple-Stream Networks for Relationship-based Captioning. *arXiv preprint arXiv:2004.12419* (2020).
- [23] Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A Shamma, et al. 2017. Visual genome: Connecting language and vision using crowdsourced dense image annotations. *International journal of computer vision* 123 (2017), 32–73.
- [24] Jie Lei, Licheng Yu, Xiaojun Yang, Yangyang Ji, Zhe Gan, Yu Cheng, Jingjing Wang, and Jingjing Liu. 2021. Learning by Aligning Visual and Language Representations. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 1217–1226.
- [25] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. 2019. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. *arXiv preprint arXiv:1910.13461* (2019).
- [26] Yu Li, Licheng Li, Bo Li, Chunyan Liu, Fan Zhang, and Yu Cheng. 2019. Entangled Transformer for Image-and-Text Matching. In *International Conference on Computer Vision (ICCV)*. IEEE, 6761–6770.
- [27] Zhen Li, Feng Zhou, Xiaoli Li, Lin Ma, and Wei Zhang. 2019. Relation-Aware Graph Attention Network for Visual Question Answering. In *International Conference on Computer Vision (ICCV)*. IEEE, 5670–5679.
- [28] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. 2014. Microsoft coco: Common objects in context. In *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13*. Springer, 740–755.
- [29] Fen Liu, Guanghui Xu, Qi Wu, Qing Du, Wei Jia, and Minghui Tan. 2020. Cascade reasoning network for text-based visual question answering. In *Proceedings of the 28th ACM International Conference on Multimedia*. 4060–4069.
- [30] Junyu Liu, Jiawei Zhang, Liqiang Nie, Yan Yan, and Li Cheng. 2019. Reasoning with Heterogeneous Graph Alignment for Video-and-Language Relationship Learning. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 9118–9127.
- [31] Ilya Loshchilov and Frank Hutter. 2017. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101* (2017).
- [32] Xiaopeng Lu, Zhen Fan, Yansen Wang, Jean Oh, and Carolyn P Rosé. 2021. Localize, group, and select: Boosting text-vqa by scene text modeling. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2631–2639.
- [33] Hyeonwoo Noh and Bohyung Han. 2016. Image Question Answering using Convolutional Neural Network with Dynamic Parameter Prediction. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 30–38.
- [34] Revant Gangi Reddy, Xilin Rui, Manling Li, Xudong Lin, Haoyang Wen, Jaemin Cho, Lifu Huang, Mohit Bansal, Avirup Sil, Shih-Fu Chang, et al. 2022. MuMuQA: Multimedia Multi-Hop News Question Answering via Cross-Media Knowledge Extraction and Grounding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 11200–11208.
- [35] Zhenwei Shao, Zhou Yu, Meng Wang, and Jun Yu. 2023. Prompting large language models with answer heuristics for knowledge-based visual question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 14974–14983.
- [36] Alon Talmor, Ori Yoran, Amnon Catav, Dan Lahav, Yizhong Wang, Akari Asai, Gabriel Ilharco, Hannaneh Hajishirzi, and Jonathan Berant. 2021. Multimodalqa: Complex question answering over text, tables and images. *arXiv preprint arXiv:2104.06039* (2021).
- [37] Damien Teney, Peter Anderson, and Xiaodong He. 2018. Tips and Tricks for Visual Question Answering: Learnings from the 2017 Challenge. In *Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. IEEE, 422–427.
- [38] Chaofan Wang, Samuel Kernan Freire, Mo Zhang, Jing Wei, Jorge Goncalves, Vassilis Kostakos, Zhanna Sarsenbayeva, Christina Schneegass, Alessandro Bozon, and Evangelos Niforatos. 2023. Safeguarding Crowdsourcing Surveys from ChatGPT with Prompt Injection. *arXiv preprint arXiv:2306.08833* (2023).
- [39] Qingqing Wang, Liqiang Xiao, Yue Lu, Yaohui Jin, and Hao He. 2021. Towards reasoning ability in scene text visual question answering. In *Proceedings of the 29th ACM International Conference on Multimedia*. 2281–2289.
- [40] Xuanyi Yang, Hanwang Huang, and James T Kwok. 2019. Hierarchical Dynamic Graph Convolutional Network for Video Question Answering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 7209–7218.
- [41] Zichao Yang, Xiaodong He, Jianfeng Gao, Li Deng, and Alexander Smola. 2015. Stacked Attention Networks for Image Question Answering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 21–29.
- [42] Zuxuan Yang, Xiaodong He, Jianfeng Gao, Li Deng, and Alexander J Smola. 2019. Making the V in VQA Matter: Elevating the Role of Image Understanding in Visual Question Answering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 6328–6337.
- [43] Zhou Yu, Jimei Yang, Youngjun Choi, Xiaohui Xiong, Alexander C Berg, and Tamara L Berg. 2018. MATTN: A Modularized Attention Network for Referring Expression Comprehension. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 1327–1335.
- [44] Zhou Yu, Jun Yu, Jianping Fan, and Dacheng Tao. 2018. Beyond Narrative Description: Generating Visual Answers from Object Scenes. *IEEE Transactions on Image Processing* 28, 1 (2018), 73–83.
- [45] Zhou Yu, Jun Yu, Jianping Fan, and Dacheng Tao. 2018. Rethinking Diversified and Discriminative Multi-Sentence Video Description Generation. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 4489–4497.

- [46] Zhou Yu, Jun Yu, Jianping Fan, and Dacheng Tao. 2019. Deep Modular Co-Attention Networks for Visual Question Answering. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 6281–6290.
- [47] Chen Zhang, Hanwang Xu, Yongzhen Huang, Jian Guo, Fei Huang, and Rongrong Ji. 2019. MAN: Moment Alignment Network for Natural Language Moment Retrieval via Iterative Graph Adjustment. In *Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 1464–1473.
- [48] Haoyuan Zhang, Ya Liu, Wanli Ouyang, Xiaogang Wang, and Dahua Lin. 2016. Yin and yang: Balancing and answering binary visual questions. In *European Conference on Computer Vision (ECCV)*. Springer, 489–505.
- [49] Minyi Zhao, Bingjia Li, Jie Wang, Wanqing Li, Wenjing Zhou, Lan Zhang, Shijie Xuyang, Zhihang Yu, Xinkun Yu, Guangze Li, et al. 2022. Towards video text visual question answering: benchmark and baseline. *Advances in Neural Information Processing Systems* 35 (2022), 35549–35562.